

LIU HENGZHI

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EDUCATION

National University of Singapore

2026 – Present

Bachelor of Computing in Computer Science (Expected to graduate in 3 years)

Singapore

Temasek Polytechnic

2023 – 2026

Diploma in Biomedical Engineering with Merit (GPA: 3.91/4.00)

Singapore

- Polytechnic Scholarship (Engineering); Course Bronze Medalist; Director's List
- Focused on embedded systems, hardware-software co-design, and applied engineering research

SKILLS

Programming Languages: Python, C++, C, Swift, SQL (MySQL)

Systems & Backend: Embedded Systems, Real-time Systems (FreeRTOS), LLM Agents, System Design Basics, Firmware Development

Hardware & Edge Systems: ESP32, Arduino, PCB Design, Sensor Integration

Tools: Git/GitHub, Linux (basic), CAD (Fusion 360 / NX / Creo), Adobe Suite

TECHNICAL PROJECTS

DTx Cognitive SaaS (ADHD Therapeutics)

Full-Stack Developer

React, FastAPI, AWS Aurora, PostgreSQL, LLM Agents

- Designed a closed-loop B2B SaaS platform for clinical ADHD cognitive assessment, integrating adaptive pacing calibrated to severity.
- Implemented IAM database token authentication via AWS SDK and local telemetry batching to optimize database connection performance.
- Designed an autonomous AI Copilot Agent via Gemini API to analyze patient telemetry, map target cognitive domains, and cache multilingual diagnostic reports.

Smart Mahjong Play-Board

Independent Systems Developer

C++, ESP32, PCB Design, Embedded Firmware

- Designed and built an ESP32-based smart Mahjong play-board prototype with custom PCB integration.
- Developed embedded firmware and hardware interfaces for real-time game interaction.

WORK EXPERIENCE

National University of Singapore (iHealthtech)

Mar 2025 – Jul 2025

Research Assistant Intern

Singapore

- Performed biological and microfluidics-based experiments, generating datasets for biomedical research studies.
- Operated laboratory instrumentation and conducted data collection, validation, and quality checks across experimental workflows.
- Analyzed experimental results and produced technical reports to support research decision-making.
- Coordinated procurement and laboratory resources to ensure uninterrupted execution of research activities.
- Maintained structured documentation and experimental records to improve workflow traceability and reproducibility.

UNIVERSITY ENGAGEMENT

- Incoming Undergraduate Student at School of Computing, National University of Singapore.
- Active engagement in computing community frameworks and prospective engineering alignment programs.

CO-CURRICULAR ACTIVITIES

WorldSkills Singapore (Additive Manufacturing)

2024

- 1st Place (Qualifying Round); demonstrated advanced CAD modeling, additive manufacturing optimization, and engineering precision design

Electronic Design & Robotics Competitions

2024

- Developed embedded logic systems and real-time control algorithms for robotics and circuit design challenges

ADDITIONAL INFORMATION

- **Leadership:** Temasek LEAD & ENGenius Excellence Programs alumnus
- **Languages:** English (Professional), Mandarin (Native)
- **Interests:** Distributed Systems, AI Infrastructure, Embedded Systems, Scientific Computing